SHANICE T. BAILEY

Environmental Learning Designer & Trainer
Climate & Ocean Science Education
Shanicetbailey.github.io | Shanice.g.bailey@gmail.com

Professional Profile

Environmental learning designer and climate scientist with 6+ years of experience designing curricula, workshops, and training programs that translate complex climate and ocean science into accessible learning experiences. Proven track record developing multi-week curricula, facilitating virtual and in-person workshops, and integrating real scientific data into education programs. Seeking remote or hybrid roles in curriculum development, climate education, and environmental training.

TEACHING/OUTREACH EXPERIENCE

PyClub Afterschool Program | Virtual

Curriculum Co-Designer & Instructor | 2020–2021

- Co-designed and implemented an **8-week afterschool curriculum** teaching Python through climate and ocean science for high school students.
- Led design of a full weekly lesson integrating real environmental datasets, data visualization, and scientific interpretation.
- Applied scaffolded, inquiry-based pedagogy to support student engagement and retention.
- Conducted **multiple formative surveys** to assess student experience; feedback highlighted increased confidence using Python and strong engagement with climate applications.
- Curriculum materials documented and made publicly available for replication.

Curriculum library: https://pyclub-cu.github.io/library.html

Coastal Ocean Environment Summer School (COESSING)

Curriculum Contributor & Instructor | Summer 2023

- Co-developed curriculum for an international summer school advancing ocean science education in West Africa.
- Helped design program schedule for virtual components and delivered lectures on research computing tools (JupyterLab, Xarray).
- Adapted advanced scientific concepts for a diverse, interdisciplinary audience.

Software Carpentry Program

Certified Instructor & Curriculum Developer | 2021–Present

- Co-developed and co-instructed remote Python courses for graduate students across scientific disciplines.

- Designed hands-on lessons, practice exercises, and live-coding demonstrations using real scientific datasets.
- Delivered instruction remotely using Zoom, Zulip, and Google Colab.

Girls Who Code
Instructor | 2019

- Designed full course syllabus and instructional materials introducing coding concepts to students.
- Delivered lectures and developed practice problems to reinforce computational thinking.

Curriculum Innovation Grant Fellow
Curriculum Developer | 2020

- Designed a technical training curriculum focused on expanding research computing skills.
- Collaborated with institutional partners to pilot the curriculum through facilitated workshops.

WORKSHOPS, TRAINING & PUBLIC ENGAGEMENT

- **GFDL Graduate School Application Workshop** (2025): Co-facilitated professional development workshop for summer interns; developed presentation materials and provided individualized feedback.
- **LEAP Train-the-Trainer Bootcamp** (2023): Provided instructional and technical support for a two-day climate data science training workshop.
- **NSBP–HGS Mentoring Program** (2020–2023): Mentored middle and high school students from underrepresented communities in STEM.
- **Public Panels & Science Outreach**: Participated in public discussions on climate science and representation in STEM.

RESEARCH & TECHNICAL EXPERIENCE

Princeton University | Geophysical Fluid Dynamics Laboratory
Postdoctoral Research Associate, Atmospheric and Oceanic Sciences **2024 – Present**

Columbia University Graduate of School of Arts & Science
Graduate Research Assistant (PhD), Earth and Environmental Sciences **2018 – 2024**

- Conducted research on ocean circulation, marine heatwaves, and climate variability using numerical models and observational datasets.
- Extensive experience working with large climate datasets and translating results for diverse audiences.
  (Publications and full research details available upon request or on website.)

EDUCATION

Columbia University Graduate of School of Arts & Science
PhD, Earth and Environmental Sciences **2024**

Dissertation: *Water mass transformation through the lens of numerical models and observations*
Advisor: Dr. Ryan P. Abernathey

MPhil, Earth and Environmental Sciences **2021**
MA, Earth and Environmental Sciences **2020**

Columbia University School of Professional Studies
Postbaccalaureate Student **2017 – 2018**

Boston University
B.A., Environmental Sciences & Wildlife- Ecology/Management **2012 – 2016**
Thesis: "Feeding Behavior of African Elephants (*Loxodonta africana)* in Tarangire-Manyara ecosystem"

PUBLICATIONS

Bailey, S. T., Jones, C. S., Abernathey, R. P.: North Atlantic Subtropical Mode Water transformation by near surface eddy mixing in high resolution climate models, *In preparation for the Journal of Geophysical Research*.

Bailey, S. T. , Drake, H. F., Gruenburg, L. K., Abernathey, R. P., & Torres, M. I. (2025). The thermodynamics of the 2023 Gulf of Mexico Marine heatwave. Geophysical Research Letters, 52, e2024GL111768, https://doi.org/10.1029/2024GL111768, 2025 .

Bailey, S. T., Jones, C. S., Abernathey, R. P., Gordon, A. L., and Yuan, X.: Water mass transformation variability in the Weddell Sea in ocean reanalyses, Ocean Sci., 19, 381–402, https://doi.org/10.5194/os-19-381-2023, 2023.

Professional Feature – ArcGIS StoryMap featured on *Estimating the Circulation and Climate of the Ocean* group's Featured Publications page. Highlights work from **Bailey, S.T., et al., 2023**.

CORE SKILLS

Curriculum design · Learning experience design · Climate & ocean science communication · Workshop facilitation · Program evaluation · Python-based data analysis · Virtual instruction · Instructional technology (Zoom, LMS, Jupyter, Google Colab)